# Supplementary Material: A Bedside Decision Support System for Disorders of Consciousness: Multi-Scale Temporal Dynamics and Multicenter Validation

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## SUPPLEMENTARY CLINICAL AND METHODOLOGICAL NOTES

### A. CRRC Cohort Recruitment and Ethics Compliance

The prospective inpatient hospital cohort was evaluated at the China Rehabilitation Research Center (CRRC / Beijing Bo'ai Hospital, Capital Medical University Rehabilitation College). Inpatients presenting with disorders of consciousness secondary to severe acute brain injury (11 intracerebral hemorrhage [ICH], 4 traumatic brain injury [TBI]) were enrolled during neurorehabilitation. Ethical approval was granted by the Medical Ethics Committee of the China Rehabilitation Research Center (Approval No. 2025-118-01; cohort recruitment and auditory paradigm screening detailed in Duan et al. [1]). Written informed surrogate consent was obtained from legally authorized representatives prior to telemetry recording in accordance with the Declaration of Helsinki. Diagnostic categorization into minimally conscious state (MCS/EMCS, $N = 6$; 5 MCS and 1 EMCS) and vegetative state / unresponsive wakefulness syndrome (VS/UWS, $N = 9$) was established through serial Coma Recovery Scale-Revised (CRS-R) assessments by certified neurorehabilitation clinicians (scores: 2–23). Recordings were conducted during stable post-acute neurorehabilitation in patients free from continuous acute ICU sedation. Telemetry was acquired from the cohort in [1]; daily medication logs were unavailable.

### B. Telemetry Acquisition and Signal Preprocessing

Bedside electroencephalography was acquired using a research-grade NeuSen NSM2 amplifier / EEGCap32-pony system configured at a native sampling rate of 1,000 Hz with active Ag/AgCl electrodes positioned according to the international 10–20 montage (19 channels; see Duan et al. [1]). Continuous recordings spanned intermediate monitoring intervals (mean $19.53 \pm 7.96$ minutes; range 7.27 to 39.07 minutes). Raw continuous signals were re-referenced to the Common Average Reference (CAR) and segmented into non-overlapping 4-second blocks (4,000 samples per block).

To prevent data leakage, artifact screening operated strictly at the individual recording level: blocks with peak-to-peak amplitude at or above $500\,\mu\mathrm{V}$ were flagged as artifactual gaps ($\mathbb{I}_{\text{adj}} = 0$). Homologous electrodes matching the discovery montage (F3, F4, C3, C4, O1, O2) were extracted.

Two subjects (CRRC_008 and CRRC_009) exhibited persistent high-impedance drift isolated to the occipital channel $O_2$. To assess clinical robustness across sensor conditions, two complementary validation protocols were evaluated:

1) **Protocol 1 (Strict Clean 6-Channel Protocol,** $N = 13$**):** Excluded the two subjects with localized $O_2$ artifacts, evaluating all records with fully intact 6-channel arrays.
2) **Protocol 2 (Full Sensitivity Protocol,** $N = 15$**):** Retained all 15 enrolled subjects under an intent-to-screen sensitivity protocol regardless of single-channel electrode impedance.

### C. Matched Cross-Center Preprocessing and Operating Points

All matched comparisons use the same feature matrix and source training records for every model. The primary matrix is $[c_{\text{global}}, -\mathrm{CV}_\theta, \rho_1, r_{\text{residual}}]$. Source-only inner three-fold cross-validation selects ridge and logistic regularization; LDA uses analytical shrinkage. Full-source feature signs are $\mathbf{s}_{\text{source}} = [+1, +1, +1, -1]^\top$. Model fitting uses source-standardized, sign-coded coordinates.

Frozen-source transfer reuses source means, scales, and signs. Target-standardized transfer computes means and scales from the unlabeled target batch while retaining source signs. CRRC target-aligned transfer uses the same target-standardized matrix and $\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$ for all models. In raw standardized coordinates, this corresponds to $\mathrm{diag}(-1, +1, +1, -1)$ applied consistently to the target inputs of source-trained supervised models. Target labels enter only the final evaluation. The direction rule is specified for the subacute clinical setting; its prospective deployment requires the rule and calibration batch to be fixed before evaluating later patients.

The primary four-feature score is the arithmetic mean; the fixed-truncation variant averages $\mathrm{clip}(s_k z_k, -1.5, 1.5)$ under identical calibration (0.9291 AUROC discovery, 0.7963 full CRRC vs. 0.9324 and 0.7778 unclipped; Table S2). Primary decisions apply source-training score medians (CRRC truncated: sensitivity 0.6667, specificity 0.7778, balanced accuracy 0.7222; Figshare source-frozen threshold: balanced accuracy 0.5645,

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TABLE S1
DE-IDENTIFIED INDIVIDUAL COHORT CHARACTERISTICS, FEATURE PROFILES, AND CROSS-CENTER BENCHMARK PERFORMANCE FOR THE CHINA REHABILITATION RESEARCH CENTER (CRRC / BEIJING BO'AI HOSPITAL) INPATIENT COHORT ($N = 15$; 15 MALES; 6 MCS [INCLUDING 1 EMCS], 9 VS/UWS; 11 ICH, 4 TBI; MEAN AGE $51.7 \pm 12.3$ YEARS). CLEAN 6-CHANNEL RECORDS ($N = 13$; 6 MCS [INCLUDING 1 EMCS], 7 VS/UWS) EXCLUDE CRRC_008 AND CRRC_009 DUE TO CONTINUOUS MECHANICAL ELECTRODE DRIFT ON $O_2$.

| Subject ID | Diagnosis | CRS-R | Lesion Laterality | Duration (s) | Retention | $c_{\text{global}}$ | $-\text{CV}_\theta$ | $\rho_1$ | $r_{\text{residual}}$ |
|---|---|---|---|---|---|---|---|---|---|
| CRRC_001 | MCS | 11 | Bilateral intact | 605.0 | 98.7% | 0.1834 | -1.3927 | +0.1118 | +0.5878 |
| CRRC_002* | MCS$^{+}$* (EMCS) | 23 | Right hemisphere | 1236.0 | 98.7% | 0.5343 | -0.3331 | +0.3201 | +0.2064 |
| CRRC_003 | MCS | 10 | Left hemisphere | 1225.0 | 97.1% | 0.8409 | -0.3682 | +0.1451 | +0.3646 |
| CRRC_004 | MCS | 7 | Left hemisphere | 1221.0 | 100.0% | 0.5993 | -0.3748 | -0.0378 | +0.1391 |
| CRRC_005 | MCS | 17 | Bilateral intact | 845.0 | 100.0% | 0.4906 | -0.6205 | +0.3850 | +0.3328 |
| CRRC_006 | MCS | 9 | Right hemisphere | 1216.0 | 97.4% | 0.5173 | -0.4568 | +0.5073 | +0.2188 |
| CRRC_007 | VS/UWS | 7 | Right hemisphere | 1896.0 | 99.8% | 0.8119 | -0.4731 | +0.0105 | +0.3776 |
| CRRC_008† | VS/UWS | 2 | Bilateral intact | 1213.0 | 100.0% | 0.7253 | -0.6384 | +0.0337 | +0.3048 |
| CRRC_009† | VS/UWS | 3 | Bilateral intact | 1209.0 | 99.7% | 0.6213 | -0.6091 | +0.0614 | +0.1660 |
| CRRC_010‡ | VS/UWS | 2 | Bilateral intact | 637.0 | 21.4% | 0.7294 | -0.4525 | +0.2401 | -0.0657 |
| CRRC_011 | VS/UWS | 7 | Bilateral intact | 1237.0 | 99.0% | 0.4154 | -0.4882 | +0.4003 | +0.3513 |
| CRRC_012 | VS/UWS | 7 | Bilateral intact | 436.0 | 99.1% | 0.5607 | -0.3266 | +0.0680 | +0.0339 |
| CRRC_013 | VS/UWS | 7 | Left hemisphere | 2344.0 | 99.5% | 0.5517 | -0.4399 | +0.1056 | +0.2702 |
| CRRC_014 | VS/UWS | 7 | Right hemisphere | 1215.0 | 86.8% | 0.7556 | -0.6032 | +0.6941 | +0.1504 |
| CRRC_015 | VS/UWS | 4 | Left hemisphere | 1040.0 | 100.0% | 0.6218 | -0.7362 | -0.0033 | +0.3001 |

*CRRC_002 met Emergence from MCS (EMCS) criteria; retained in conscious class (MCS$^{+}$) for binary triage. †Mechanical $O_2$ electrode drift; excluded in Protocol 1 ($N = 13$), retained in Protocol 2 ($N = 15$). ‡21.4% retention due to motion; retained under intent-to-treat. Telemetry: NeuSen NSM2 / EEGCap32-pony (1000 Hz, 4-s epochs, CAR, F3, F4, C3, C4, O1, O2; IRB No. 2025-118-01 [1]). Model comparisons across protocols appear in Table S2; operating thresholds are detailed in Note C.

TABLE S2
MATCHED-INPUT MODEL COMPARISONS (AUROC). ALL METHODS SHARE FEATURE COLUMNS, SOURCE-ONLY PARAMETER SELECTION, SCALING, DIRECTION CODING, AND SOURCE-TRAINING MEDIAN THRESHOLDS WITHIN EACH PROTOCOL. TRUNCATION IS A FIXED SCORING-RULE VARIANT ($c = 1.5$). TARGET STANDARDIZATION AND DIRECTION ALIGNMENT ARE UNLABELED BATCH ADAPTATIONS.

| Cohort and protocol | Linear | Clipped | Ridge | Logistic | LDA |
|---|---|---|---|---|---|
| Mendeley, 25 outer folds | 0.9324 | 0.9291 | 0.8813 | 0.8964 | 0.8769 |
| CRRC 15, source frozen | 0.5741 | 0.5741 | 0.5556 | 0.5556 | 0.5556 |
| CRRC 15, target standardized | 0.5000 | 0.5000 | 0.5185 | 0.5185 | 0.5370 |
| CRRC 15, target aligned | 0.7778 | 0.7963 | 0.7407 | 0.7593 | 0.7593 |
| CRRC 13, target aligned | 0.7143 | 0.7143 | 0.7143 | 0.6905 | 0.6905 |
| Figshare 50, $D_1$, source frozen | 0.7691 | 0.7606 | 0.7759 | 0.7674 | 0.7742 |
| Figshare 50, $D_1$, target standardized | 0.7657 | 0.7606 | 0.7708 | 0.7589 | 0.7691 |

sensitivity 1.0000, specificity 0.1290). All transforms, traces, and input hashes are archived in results/matched_model_comparison_v1/; paired uncertainty applies 10,000 subject-level Dirichlet draws.

### *D. Section S-CDSS: Bedside Streamlined Implementation & Latency Profiling*

To explore implementation feasibility, the four-feature temporal dynamic consensus pipeline was implemented as a streaming prototype (docs/bedside_cdss_demo.html, animation in docs/bedside_playback_demo.gif). The architecture accepts consecutive non-overlapping 4-s telemetry epochs ($f_s = 256\,\text{Hz}$), subjecting each block to artifact gating ($< 500\,\mu\text{V}$). Temporal continuity tracking ensures that artifact gaps break physical adjacency ($\mathbb{I}_{\text{adj}} = 0$); feature estimation is gated until $\geq 10$ physically adjacent clean pairs accumulate. Data with $< 10$ adjacent pairs yield `INSUFFICIENT_DATA`. Source-frozen standardized coordinates ($z_k = s_k(x_k - \mu_k)/\sigma_k$) use archived thresholds ($\theta_0 = 0.0451$ for the raw score and $\theta_{1.5} = 0.0725$ for the clipped score), yielding illustrative candidate prioritization rather than definitive clinical referrals.

Cached-power replay of two source-training records (5,400 epochs per subject; P25 [VS] and P36 [MCS]) showed numerical agreement with offline batch processing ($|\Delta_{\text{feature}}| \leq 2.77 \times 10^{-13}$, scoring error 0.000; Table S3). Cached-power replay timings were: cached power ingestion mean 0.0098 ms (p95 0.011 ms); four-feature extraction mean 2.13 ms (p95 3.11 ms); consensus scoring mean 0.011 ms; and total evaluated update mean 2.15 ms (p95 3.13 ms), well below the 4,000 ms inter-epoch window. Raw EEG preprocessing and clinical decision latency were not measured; no single-core or thread restriction was imposed. Table values average the two subject-level summaries (27 evaluated updates each), including their individual p95 values. Complete trajectories and test records are archived in results/bedside_playback_demo/.

## REFERENCES

[1] F. Duan *et al.*, "Auditory paradigm-dependent EEG feature screening and representation for distinguishing consciousness states in male patients with disorders of consciousness," *Frontiers in Neurology*, vol. 17, p. 1927313, 2026.

TABLE S3
CACHED-POWER REPLAY TIMINGS AND VERIFICATION (5,400 EPOCHS PER RECORD; TWO RECORDS)

| Processing Phase | Calls/record | Mean (ms) | Mean p95 (ms) | Max (ms) |
|---|---|---|---|---|
| Cached Power Ingest | 5,373 | 0.0098 | 0.0111 | 0.7090 |
| Feature Extraction | 27 | 2.1323 | 3.1117 | 3.9057 |
| Consensus Scoring | 27 | 0.0110 | 0.0123 | 0.0164 |
| Evaluated Step Total | 27 | 2.1542 | 3.1330 | 3.9261 |

Offline Consistency: Max feature $|\Delta| \leq 2.77 \times 10^{-13}$; Score error: 0.000.